

**Syllabus (Linear Algebra)**  
**of Mid Term Spring – 2023**

**Matrix and Determinants (2.5 sets)**

**Definition:**

Different types of matrix, Transpose of matrix, Inverse of matrix, Rank of a matrix, Determinant.

**Problem Types:**

**(Multiplication, Transpose, Symmetric/Skew Symmetric, Orthogonal, Involutory)**

(1) Let  $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 1 & 3 & 2 & 5 \\ 2 & 1 & -1 & 3 \end{bmatrix}$ . Calculate the product AB.

(2) If  $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$  Prove that  $A^3 + A^2 - 21A - 54I = 0$ , Where I is the identity matrix of order  $3 \times 3$ .

(3) If  $A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 5 & -1 \\ 2 & 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} -1 & 5 & 3 \\ 7 & -2 & 1 \\ 2 & 0 & -3 \end{bmatrix}$  then prove that  $(AB)^T = B^T A^T$ .

(4) Find the symmetric and skew symmetric parts of the matrix  $A = \begin{bmatrix} 1 & 2 & 4 \\ 6 & 8 & 1 \\ 3 & 5 & 7 \end{bmatrix}$ .

(5) Prove that the matrix  $A = \frac{1}{6} \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & -5 & 1 & 1 \\ 3 & 1 & 1 & -5 \\ 3 & 1 & -5 & 1 \end{bmatrix}$  is orthogonal.

(6) Show that the matrix  $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$  is involutory.

**Determining Inverse –**

(1) If  $A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$  find  $A^{-1}B$ .

(2) Find the inverse of matrix  $A = \begin{bmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{bmatrix}$ .

(3) Find the inverse of the matrix  $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix}$ .

(4) Find the inverse of the matrix  $A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$  using RREF form.

(5) Find the inverse of the following matrix by using reduced row echelon form:

$$A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix}$$

(6) If  $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$ , then find the values of  $A^{-1}B$  and  $AB^{-1}$ .

### Determining Rank of Matrix:

(1) Find the rank of the matrix  $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ .

(2) Find the rank of the matrix  $A = \begin{bmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{bmatrix}$

(3) Determine the rank of the following matrix:  $B = \begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ ,

$$C = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

### Solving System of Linear Equations:

(1) Solve the following SLE with the help of inverse of matrices:

|                  |                    |
|------------------|--------------------|
| (i)              | (ii)               |
| $x + y + z = 6$  | $3x + 2y - z = 20$ |
| $x - y + z = 2$  | $2x + 3y - 3z = 7$ |
| $2x + y - z = 1$ | $x - y + 6z = 41$  |
| Sol: (1,2,3)     | Sol: (5,6,7)       |

(2) Use Gauss-elimination method to solve the following SLEs

(a)  $x + y + z = 6$  ;  $2x + 3y + 4z = 20$  ;  $3x - 2y + z = 2$

(b)  $x + 2y + 3z = 5$  ;  $2x - y + z = 5$  ;  $4x + 2y - 3z = 5$

(c)  $x + y + z + t = 2$  ;  $2x + 3y + 4z + 5t = 9$  ;  $3x - 2y + z + 2t = 9$  ;  $x + 2y - 2z + 3t = 5$

### Vectors (1.5 Sets)

**Definition:** Vector Space, Linear Dependence/ Independence, Eigen value/ Eigen Vectors

Problems:

**Determining Linearly dependent/independent:** Given in Lecture note + Exercises – (Page – 271):  
1, 2, 3, 6, 15

**Determine Eigen Value/Eigen vectors:** Solved and given problems in the class (I don't have the book with me, so couldn't include problem numbers. Find yourself).